

Day 5: Case Study Practical : Work on a scenario involving multiple hardware issues.

Tasks:-

1. A user reports their laptop won't power on even when plugged in. List 5 troubleshooting steps you would take to diagnose the cause, and for each step, explain what you are checking for.

Ans.:-

steps I would take to diagnose why a laptop won't power on, even when plugged in:-

1. Check the power source and AC adapter :-

- What I'm checking for: Whether the laptop is actually receiving power.
- Verify that the wall outlet works by plugging in another device.
- Inspect the charger for frayed cables, bent connectors, or damage.
- If possible, test with a known-good compatible charger to rule out a faulty power adapter.

2. Inspect the charging port and power indicators :-

- What I'm checking for: Signs that power is reaching the laptop.
- Look for charging LEDs or status lights when the adapter is connected.
- Check if the charging port is loose, damaged, or obstructed by debris, which could prevent a proper electrical connection.

3. Perform a power reset (hard reset)

- What I'm checking for: Whether the laptop is stuck due to residual electrical charge or a firmware issue.
- Disconnect the charger, remove the battery (if removable), hold the power button for 15–30 seconds, then reconnect the charger and try powering it on again.
- If the laptop starts afterward, the issue was likely related to a temporary power state rather than failed hardware.

4. Test the battery :-

- What I'm checking for: Whether a failed battery is preventing startup.
- If the battery is removable, remove it and try powering on using only the AC adapter.
- If the laptop powers on without the battery, the battery may be defective. If it still does not power on, the problem likely lies elsewhere.

5. Check for hardware or motherboard issues :-

- What I'm checking for: Evidence of internal hardware failure.
- Listen for fan noise, beeps, or observe keyboard backlights when pressing the power button.
- Disconnect external devices (USB drives, docking stations, etc.) that could interfere with startup.
- If there are no signs of power despite a working charger and outlet, the issue may involve the motherboard, DC power jack, RAM, or another internal component requiring further hardware diagnostics or repair.

**2. You receive a helpdesk ticket: 'My monitor is showing a blank screen, but the CPU seems to be running.' Write a step-by-step troubleshooting flow you would follow to identify and resolve the issue.

Hint: Consider checking both hardware connections and basic display settings.**

Ans.:-

Helpdesk Ticket: My monitor is showing a blank screen, but the CPU seems to be running.

Step 1: Verify the Monitor Has Power

Check:

- Is the monitor's power LED on?
- Is the power cable securely connected?
- Is the wall outlet or power strip working?

Action:

- Turn on the monitor.
- Try a different power outlet if needed.

Step 2: Check the Display Cable

Check:

- Ensure the HDMI, DisplayPort, DVI, or VGA cable is firmly connected to both the monitor and the computer.
- Look for bent pins or cable damage.

Action:

- Reseat the cable.
- Try another display cable if available.

Step 3: Verify the Correct Input Source

Check:

- Is the monitor set to the correct input (HDMI, DisplayPort, VGA, etc.)?

Action:

- Use the monitor's Input/Source button to select the correct input.

Step 4: Confirm the Cable Is Connected to the Correct Video Port

Check:

- If the PC has a dedicated graphics card, make sure the monitor is connected to the graphics card output—not the motherboard video port (unless integrated graphics are being used).

Action:

- Move the cable to the correct video output if necessary.

Step 5: Restart the Computer

Check:

- Does the monitor display anything during startup (manufacturer logo or BIOS screen)?

Action:

- Restart the PC and observe whether any image appears.

Step 6: Test the Monitor

Check:

- Connect the monitor to another computer or laptop.
- Alternatively, connect a different monitor to the PC.

Action:

- If the monitor works elsewhere, the problem is likely with the PC.
- If it doesn't, the monitor may be faulty.

Step 7: Check Display Settings

Check:

- If Windows loads but the screen stays blank, the display settings may be incorrect.

Action:

- Press Windows + P and select Duplicate or PC screen only.
- If using multiple monitors, disconnect the secondary display and test again.

Step 8: Check Internal Hardware**Check:**

- Power off the PC.
- Reseat the graphics card (if installed).
- Reseat the RAM modules.
- Ensure all power connectors are secure.

Action:

- Boot the system again after reseating components.

Step 9: Listen for POST Beeps or Error Indicators**Check:**

- Listen for beep codes or observe motherboard diagnostic LEDs.

Action:

- Use the motherboard manual to interpret any error codes and identify faulty hardware.

Step 10: Update or Reinstall Graphics Driver (If Display Works in Safe Mode)**Check:**

- Boot into Safe Mode if possible.

Action:

- Update or reinstall the graphics driver.
- Restart the computer normally.

3. A user complains that their wireless mouse is unresponsive, but the keyboard works fine. Describe how you would isolate whether the issue is with the mouse, the USB port, or the system settings.

Ans.:-

Troubleshooting: Wireless Mouse Is Unresponsive but Keyboard Works

Step 1: Check the Mouse

What to check:

- Ensure the mouse is switched ON.
- Check if the batteries are discharged.
- Look for any LED or optical sensor light.

Purpose:

- Determines whether the mouse itself has power.

Step 2: Check the USB Receiver

What to check:

- Verify the wireless USB receiver (dongle) is fully inserted.
- Remove and reconnect the receiver.

Purpose:

- Confirms the receiver has a proper connection to the computer.

Step 3: Test a Different USB Port

What to check:

- Plug the USB receiver into another USB port (preferably a USB 2.0 port if available).

Purpose:

- Determines whether the original USB port is faulty.

Result:

- If the mouse works in another port, the original USB port is likely defective.

Step 4: Test the Mouse on Another Computer

What to check:

- Connect the mouse and its USB receiver to a different computer.

Purpose:

- Determines whether the mouse hardware is functioning.

Result:

- If it doesn't work on another computer, the mouse or receiver is likely faulty.
- If it works, the issue is with the original computer.

Step 5: Check Device Manager

What to check:

- Open Device Manager (devmgmt.msc).
- Look under Mice and other pointing devices and Universal Serial Bus controllers for warning icons.

Purpose:

- Identifies missing, disabled, or corrupted drivers.

Action:

- Update, enable, or reinstall the mouse driver if needed.

Step 6: Check System Settings

What to check:

- Open Settings → Bluetooth & devices → Mouse.
- Ensure mouse settings are normal and no accessibility options (such as Mouse Keys) are interfering.

Purpose:

- Verifies that Windows settings are not preventing normal mouse operation.

Step 7: Restart the Computer

What to check:

- Reboot the system and test the mouse again.

Purpose:

- Clears temporary software or driver issues.

**4. Imagine you are working on a helpdesk team for a gaming cafe. One system is overheating and shutting down during use. List 3 possible hardware causes and describe how you would check each one.

Hint: Think about fans, dust, and thermal paste.**

Ans.:-

Possible Hardware Cause	How to Check It	Possible Solution
1. Faulty or Non-Working Cooling Fan	Open the case and check whether the CPU fan, GPU fan, and case fans are spinning when the PC is powered on. Listen for unusual noises and ensure fan cables are securely connected to the motherboard or power supply.	Reconnect loose cables, clean the fan, or replace the faulty fan if it does not spin properly.
2. Dust Buildup	Inspect the inside of the PC for dust on the CPU heatsink, GPU, fans, air vents, and filters. Excessive dust blocks airflow and traps heat.	Use compressed air to clean the fans, heatsinks, and vents. Ensure there is adequate airflow inside the case.
3. Dried or Improperly Applied Thermal Paste	Monitor CPU temperatures in the BIOS or with a hardware monitoring tool. If temperatures rise quickly even when the CPU fan is working, remove the CPU cooler and inspect the thermal paste.	Clean off the old thermal paste with isopropyl alcohol, apply a fresh layer of high-quality thermal paste, and reinstall the cooler securely.